Md. Ridwan Hossain Talukder

Ph.D. Candidate, Computer Science, George Mason University
Research Focus: Neuro-Symbolic Task and Motion Planning for Long-Lived Robots in Human-Shared Spaces
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RESEARCH INTERESTS

Neuro-symbolic planning, task and motion planning, learning for planning, long-horizon decision-making, multi-agent and human-robot shared environments.

EDUCATION

Ph.D. Computer Science, George Mason University, Fairfax, VA *Expected May 2027*
Advisor: Gregory J. Stein | Dissertation: *Courteous and Anticipatory Task and Motion Planning for Long-Lived Service Robots*

M.S. Computer Science, George Mason University, Fairfax, VA (GPA: 3.97/4.0)
B.S. Computer Science and Engineering, University of Dhaka, Bangladesh

PUBLICATIONS

Peer-Reviewed

[C2] **MRH Talukder**, RI Arnob, GJ Stein. Anticipatory Planning for Performant Long-Lived Robot in Large-Scale Home-Like Environments. *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[C1] R Dhakal, **MRH Talukder**, GJ Stein. Anticipatory Planning: Improving Long-Lived Planning by Estimating Expected Cost of Future Tasks. *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.

Under Review

[S1] **MRH Talukder**, E Phillips, GJ Stein. Courteous Anticipation: Improving Long-Lived Task Planning in Shared Environments. *Submitted to IEEE ICRA*, 2027.

RESEARCH EXPERIENCE

Graduate Research Assistant, RAIL Lab, George Mason University Jan 2022 – Present

- Developed anticipatory planning algorithms that let long-lived robots reason over future task distributions, reducing long-horizon execution cost by ≈**40%** over myopic baselines.
- Proposed neuro-symbolic cost estimation integrating graph neural networks over scene graphs with PDDL planning, scaling to environments with **10+** rooms and **100+** objects.
- Introduced courteous anticipation for planning in human-shared spaces; evaluated across **50,000** planning episodes in ProcTHOR against myopic and non-courteous baselines.

FELLOWSHIPS & AWARDS

Predoctoral Fellow, Institute of Digital Innovation (IDIA), GMU Aug 2024 – Present
NSF Research Trainee, Center for Advancing Systems Science (CASSBI), GMU Jun 2021 – May 2022
Outstanding Graduate Teaching Assistant Award, Volgenau School of Engineering, GMU May 2023

TEACHING

Graduate Teaching Assistant, George Mason University Jan 2021 – May 2024
6 semesters across Artificial Intelligence, Data Structures, Data Mining, Software Engineering, and Software Testing. Led a lab section and exam review sessions; held weekly office hours.

SERVICE & LEADERSHIP

Bangladeshi Graduate Student Association, GMU Apr 2021 – Apr 2023
President (2022–23); Director of Public Relations (2021–22)

SKILLS

Planning: PDDL / Temporal PDDL, FastDownward, PDDLStream, MA-STRIPS, multi-agent planning
Learning & Simulation: PyTorch, GNNs, ROS2, ProcTHOR / AI2-THOR, Isaac Sim, MuJoCo, PyBullet, Gazebo
Languages: Python, C++ **Prior industry:** Software Engineer (2017–2020); Founder & CTO, Khelo.io (2020)